

Technology Stack

Date	03 JULY 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Provides a responsive, user-friendly, and interactive web interface for entering HDI indicators, viewing dashboards, charts, and prediction results across different devices.
2	Backend / Server-Side	Python, Flask	Flask provides a lightweight and efficient backend framework to process user requests, communicate with the machine learning model, and return HDI predictions quickly.
3	Database / Data Storage	HDI Dataset (CSV), Pandas DataFrames	The HDI dataset is stored in CSV format and managed using Pandas, enabling efficient preprocessing, analysis, and model training.
4	Cloud / Hosting / Deployment	Flask Local Server / Render (or Local Deployment)	Enables easy deployment, testing, and demonstration of the web application while allowing future migration to cloud platforms if required.

5	Version Control & CI/CD	Git, GitHub	Git and GitHub provide version control, team collaboration, source code management, and project tracking throughout development.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib	Scikit-learn is used for machine learning model development, Pandas and NumPy for data preprocessing, Matplotlib and Seaborn for visualizations, and Joblib for saving and loading the trained model.